HISTOLOGY · MCQ SET

Large Intestine & Anal Canal

SGL 3
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GIT Block

Histology GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 4 case-based. Attempt every question before checking the answer key on the last page.

- 4 Easy
- 10 Medium
- 6 Hard

1. EASY

Which feature is absent in the large intestine but present in the small intestine, and is the fastest way to distinguish the two on a slide?

A. Crypts of Lieberkühn
B. Goblet cells
C. Submucosa
D. Villi
E. Muscularis mucosae

2. EASY

Teniae coli are:

A. Lymphoid nodules unique to the appendix
B. A continuous sheet of outer longitudinal muscle found throughout the colon and rectum
C. Three bands of outer longitudinal muscle, absent in the appendix and rectum
D. The epithelial lining of the anal canal
E. A feature exclusive to the small intestine

3. EASY

The dentate (pectinate) line marks the transition between which two epithelial types?

A. Stratified squamous to transitional epithelium
B. Simple squamous to simple columnar
C. Simple columnar to non-keratinized stratified squamous
D. Simple cuboidal to stratified columnar
E. Pseudostratified to simple columnar

4. EASY

Which anal sphincter is composed of skeletal muscle and is under voluntary control?

A. Internal anal sphincter
B. Neither is skeletal muscle
C. External anal sphincter
D. Both are skeletal muscle
E. Internal sphincter, in the lower third only

5. MEDIUM CASE

A colonic biopsy shows long straight crypts, absent villi, and abundant goblet cells.

Which further feature would confirm this is from the transverse colon rather than the rectum?

A. Presence of Brunner's glands
B. Presence of a dentate line
C. Presence of teniae coli
D. Presence of villi
E. Presence of Peyer's patches

6. MEDIUM

A biopsy from the appendix would be expected to show all of the following EXCEPT:

A. Fewer, shorter glands than colon

B. Narrow, irregular lumen

C. Teniae coli

D. Abundant lymphoid nodules

E. A 2-layer muscularis externa (inner circular + outer longitudinal)

7. MEDIUM

Which region of the large intestine lacks teniae coli, with the outer longitudinal muscle forming a complete sheet again, similar to elsewhere in the GI tract?

A. Ascending colon

B. Descending colon

C. Rectum

D. Sigmoid colon

E. Transverse colon

8. MEDIUM

Which segments of the large intestine are intraperitoneal and therefore covered by serosa?

A. Ascending and descending colon

B. The entire large intestine uniformly

C. Rectum only

D. Anal canal only

E. Cecum, transverse colon, and sigmoid colon

9. MEDIUM

The lowest 8–10 mm of the anal canal is histologically distinct because it is:

A. Lined by plicae circulares

B. Marked by teniae coli

C. True skin, with hair follicles, sebaceous glands, and sweat glands

D. Lined by simple columnar epithelium with Peyer's patches

E. Covered by a serosa

10. MEDIUM

Anal columns of Morgagni are located in which zone of the anal canal, and are lined by which epithelium?

A. Upper anal canal (~15 mm); simple columnar, like the rectum

B. Throughout the entire canal uniformly; simple cuboidal

C. Lowest 8–10 mm; keratinized stratified squamous

D. Middle zone only; transitional epithelium

E. Only at the dentate line itself; parakeratotic squamous

11. MEDIUM

Which venous plexus lies above the dentate line, relevant to internal haemorrhoid formation?

A. Superior mesenteric plexus

B. Portal venous plexus directly

C. External hemorrhoidal plexus

D. Internal hemorrhoidal plexus

E. Pampiniform plexus

12. MEDIUM

Which of the following crypt/mucosal cell types was explicitly noted as NOT mentioned for the large intestine in this lecture, to avoid overstating what was taught?

A. Goblet and absorptive cells
B. Paneth and M cells
C. Endocrine cells
D. Enteroendocrine cells only
E. Stem cells

13. MEDIUM

Compared to the small intestine, the large intestine's crypts (Crypts of Lieberkühn) are:

A. Shorter than small intestinal crypts
B. Replaced by villi
C. Absent entirely
D. Identical in every respect
E. Longer and straighter than small intestinal crypts

14. MEDIUM

Which large intestinal region is a blind evagination and the first part of the large intestine encountered after the ileocecal valve?

A. Sigmoid colon
B. Anal canal
C. Descending colon
D. Rectum
E. Cecum (with the appendix)

15. HARD CASE

A pathologist reviews two colonic segments. Segment A shows a normal 3-band longitudinal muscle pattern; Segment B shows a complete circumferential sheet of longitudinal muscle instead of discrete bands, with otherwise typical colonic crypt architecture and goblet cells.

Which segment is most likely the rectum, and what is the key discriminating feature?

A. Segment B, because the rectum uniquely lacks teniae coli and instead has a complete, non-banded outer longitudinal muscle sheet
B. Segment A, because a complete muscle sheet only occurs in the appendix
C. Neither — both patterns are equally consistent with any colonic segment
D. Segment B, because the sheet pattern indicates small intestine
E. Segment A, because teniae coli are a rectal-specific feature

16. HARD

A biopsy from an unspecified region shows abundant lymphoid nodules, a narrow irregular lumen, fewer and shorter glands than typical colon, and a standard 2-layer muscularis externa without teniae coli. Which segment is this, and which feature distinguishes it from typical colon beyond the missing teniae coli?

A. Cecum; distinguished by prominent teniae coli
B. Sigmoid colon; distinguished by absent goblet cells
C. Rectum; distinguished by its complete muscle sheet alone
D. Ascending colon; distinguished by presence of villi
E. Appendix; distinguished by its narrow irregular lumen and heavy lymphoid infiltration, unlike the wider, less lymphoid-dense typical colon

17. HARD CASE

A surgical specimen from the anal region shows an abrupt transition from simple columnar epithelium to non-keratinized stratified squamous epithelium, with anal valves visible joining the bases of longitudinal mucosal folds just proximal to the transition.

Which structure has been identified, and what is its clinical/anatomical significance?

A. The linea alba, a fascial landmark unrelated to epithelial transition
B. The ileocecal valve, marking the small-to-large bowel transition
C. The anal verge, marking the boundary with perianal skin
D. The anorectal ring, marking the boundary of the external sphincter
E. The dentate (pectinate) line, formed by anal valves joining the bases of the anal columns of Morgagni, dividing the canal into upper two-thirds and lower one-third with different embryological origins

18. HARD

A patient develops internal haemorrhoids (dilation of the venous plexus above the dentate line) but reports minimal pain, while external haemorrhoids near the anal verge are markedly painful. What histological/anatomical basis best explains this difference in sensation?

A. External haemorrhoids arise from the internal anal sphincter exclusively
B. Both plexuses share identical innervation, so the difference is coincidental
C. Internal haemorrhoids never actually involve mucosa
D. The region above the dentate line (upper anal canal/rectum) is lined by visceral-type mucosa with relatively less somatic sensory innervation, whereas the lowest anal canal is true skin with dense somatic sensory supply
E. The dentate line has no relevance to pain sensation

19. HARD

Which combination of findings would most strongly support a diagnosis of 'colon' rather than 'small intestine' on an unlabelled biopsy, integrating multiple discriminators from this lecture?

A. Presence of a wide lumen alone, without other features
B. Presence of Brunner's glands in the submucosa
C. Presence of villi with numerous goblet cells
D. Absence of villi, long straight crypts, numerous goblet cells, and abundant lymphoid nodules — the combined discriminator set for large intestine versus small intestine
E. Presence of plicae circulares with a narrow lumen

20. HARD CASE

A trainee is asked to localise a biopsy to either the ascending colon or the rectum, given that both show typical colonic crypt architecture, goblet cells, and no villi.

Which additional finding would most reliably differentiate ascending colon from rectum?

A. Goblet cell density, since ascending colon has far more goblet cells than rectum
B. Muscularis mucosae thickness, which differs markedly between the two
C. Presence of Peyer's patches in the ascending colon only
D. Presence of teniae coli (ascending colon) versus their absence with a complete longitudinal muscle sheet (rectum), combined with serosa (ascending colon, since it is retroperitoneal with both adventitia and serosa) versus adventitia-only covering (rectum)
E. Crypt length, since rectal crypts are shorter than ascending colon crypts

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	**D**	Villi
2	**C**	Three bands of outer longitudinal muscle, absent in the appendix and rectum
3	**C**	Simple columnar to non-keratinized stratified squamous
4	**C**	External anal sphincter
5	**C**	Presence of teniae coli
6	**C**	Teniae coli
7	**C**	Rectum
8	**E**	Cecum, transverse colon, and sigmoid colon
9	**C**	True skin, with hair follicles, sebaceous glands, and sweat glands
10	**A**	Upper anal canal (~15 mm); simple columnar, like the rectum
11	**D**	Internal hemorrhoidal plexus
12	**B**	Paneth and M cells
13	**E**	Longer and straighter than small intestinal crypts
14	**E**	Cecum (with the appendix)
15	**A**	Segment B, because the rectum uniquely lacks teniae coli and instead has a complete, non-banded outer longitudinal muscle sheet
16	**E**	Appendix; distinguished by its narrow irregular lumen and heavy lymphoid infiltration, unlike the wider, less lymphoid-dense typical colon
17	**E**	The dentate (pectinate) line, formed by anal valves joining the bases of the anal columns of Morgagni, dividing the canal into upper two-thirds and lower one-third with different embryological origins
18	**D**	The region above the dentate line (upper anal canal/rectum) is lined by visceral-type mucosa with relatively less somatic sensory innervation, whereas the lowest anal canal is true skin with dense somatic sensory supply
19	**D**	Absence of villi, long straight crypts, numerous goblet cells, and abundant lymphoid nodules — the combined discriminator set for large intestine versus small intestine
20	**D**	Presence of teniae coli (ascending colon) versus their absence with a complete longitudinal muscle sheet (rectum), combined with serosa (ascending colon, since it is retroperitoneal with both adventitia and serosa) versus adventitia-only covering (rectum)